BEM 103

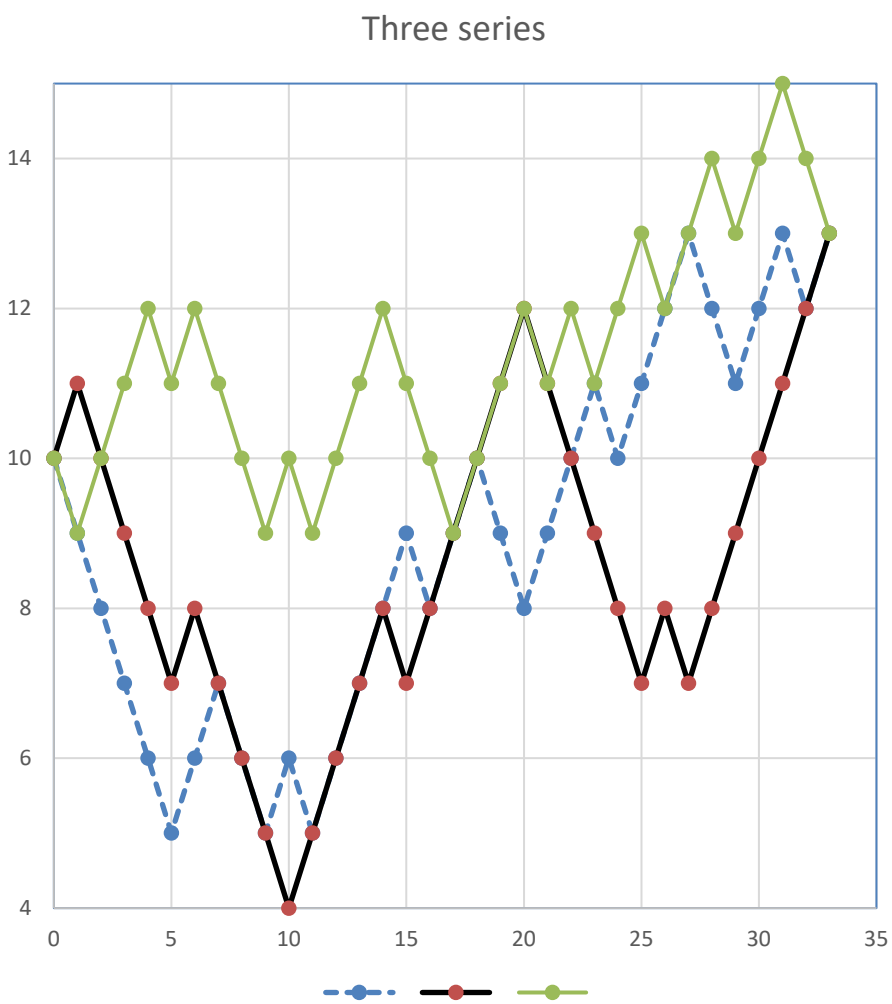
Class 09
The Portfolio approach to risk

- More than one security out there and returns not perfectly correlated
- Efficient frontier with two assets
- Dominated portfolios
- Risk free asset and the tangent portfolio
- Beyond 2 assets
- Interpretation
  - Frontier and mean-variance.
  - Why not insurance contracts?

Key concepts

- Mean variance
- Optimal portfolio
- Efficient frontier
- Dominated portfolios
- Efficient frontier with a riskless asset
- Short selling

Investment with risk



	S1	S2	S3
R	3.00	3.00	3.00
S1	1.02	0.02	0.14
S2		1.02	0.02
S3			1.02

Mean Variance says

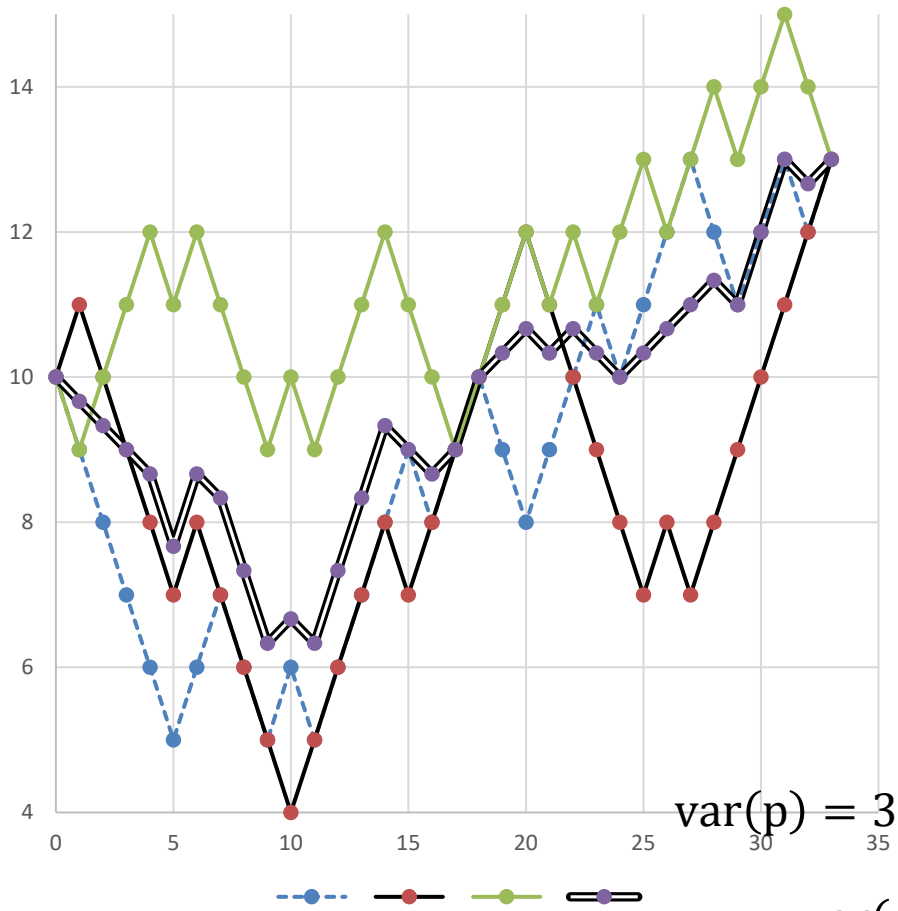
P=r-bvar

P=3 -b(1.02)

Same price for all series

Investment with risk

Add the equal weight portfolio



	S1	S2	S3	Portfolio
R	3.00	3.00	3.00	3.00
S1	1.02	0.02	0.14	0.38
S2		1.02	0.02	
S3			1.02	

$$\mathrm{var(p)} = \sum_{i=1}^{n}\sum_{j=1}^{n} q_i q_i \sigma_{ii}$$

$$\mathrm{var(p)} = 3\,\frac{1}{3}\frac{1}{3}1.02 + 2\,\frac{1}{3}\frac{1}{3}\;0.02 + 2\,\frac{1}{3}\frac{1}{3}0.14 + 2\,\frac{1}{3}\frac{1}{3}\;0.02$$

$$\mathrm{var(p)} = \frac{3}{9}1.02 + \frac{2}{9}\;0.02 + \frac{2}{9}0.14 + \frac{2}{9}\;0.02$$

$$P = 3\; - b(.38)$$

Lessons from the example

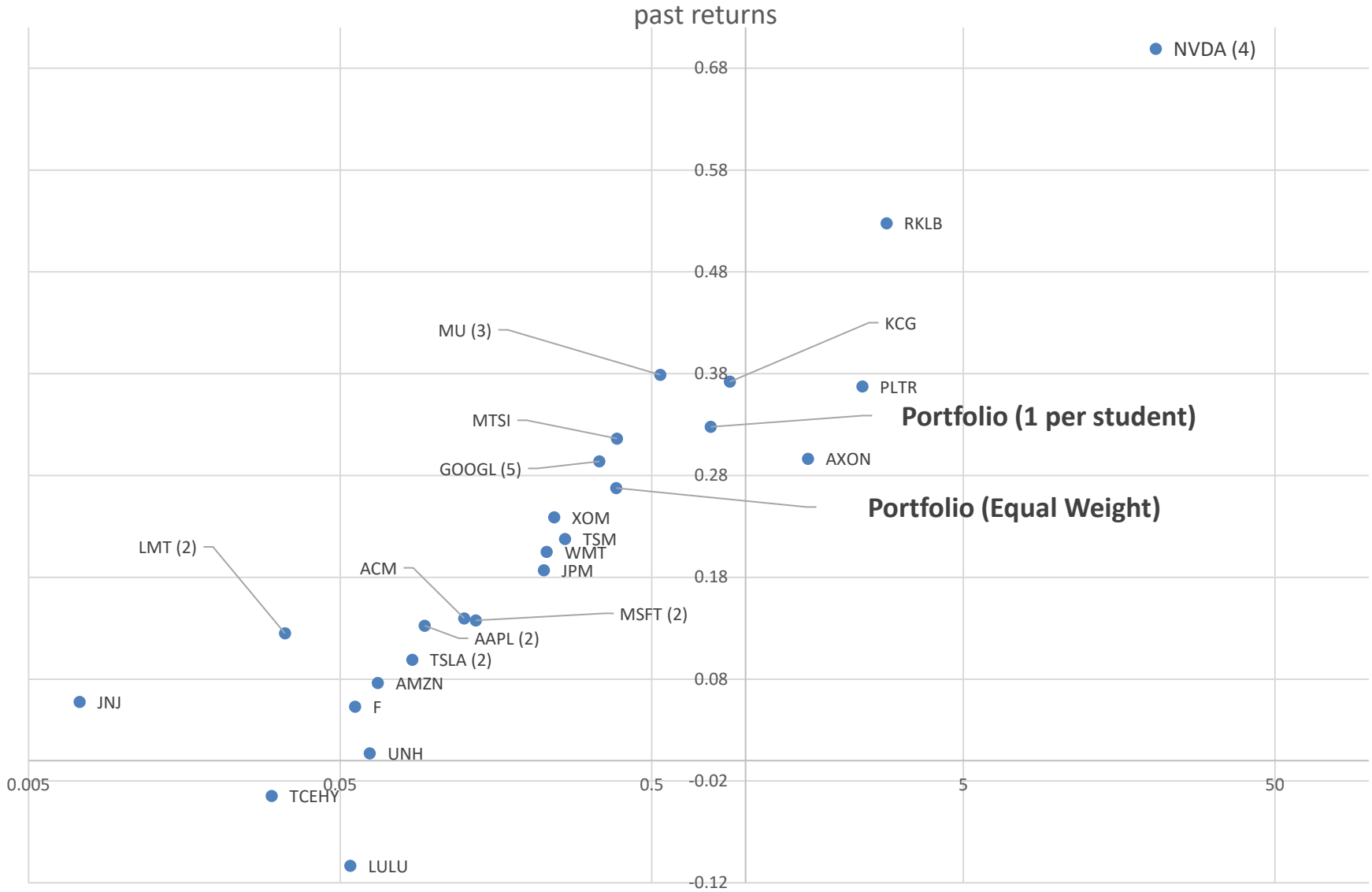
- The portfolios has a better mean-variance property than the assets
  - Return are linear combinations if returns
  - Variances
  - So the price of the assets are wrong.
- Need to investigate portfolios taking prices as fixed.
  - Optimal portfolios
    - Markowitz 1952 (Nobel 1990).
    - Sharpe 1966 (Nobel 1990).
  - Implication for asset prices
    - CAPM
    - APT

Portfolio choices

- If risk-neutral only 1 security (the one that the investor thinks has the highest expected value)
- If not risk-neutral
  - Then willing to give up some return for lower volatility, variance, risk
- This is where portfolio choices come in.
- Explore these with publicly traded securities
  - because the quantitative part is easy
  - Same ideas apply in very wide variety of contexts

Last weeks investment

(5 year past returns monthly)



Mean and variance of a portfolio

- Let w_i be the weight of asset x_i in the portfolio (its share of total value)
- The mean return of the portfolio is the weighted average of the individual returns

$$E(w_1x_1, \dots, w_nx_n) = \sum_{1}^{n} w_i E(x_i)$$

- The variance of a portfolio is the weighted sum of the variances and covariances.

$$Var(w_1x_1, \dots, w_nx_n) = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij}$$

- Objective 1 find vector W st for a given return, variance is minimized.
- Objective 1' find vector W st for a given variance return is maximized.

- Turns out unique set of W vectors that trace a mean variance frontier.

For two assets.

$$E(w_1x_1, w_2x_2) = w_1E(x_1) + w_2E(x_2)$$

$$Var(w_1x_1, w_2x_2) = w_1^2\sigma_{11} + 2w_1w_2\sigma_{12} + w_2^2\sigma_{22}$$

- Return is weigted average (range $E(x_1)$, $E(x_2)$)
- Because the weights are less than 1, so their squares are small.
- Obvious: if covariance negative your portfolio will have lower variance than either assets.
- Also true in other cases because $\sigma_{12} < max\{\sigma_{11}, \sigma_{22}\}$

Optimal portfolio

- All you need to decide are the weights
  - (e.g potatoes and rye or Amazon vs Google, 3 year T-Bill vs Wells Fargo Stock).
- Problem chose w_1 and w_2 to minimize .
- Min $Var(w_1x_1 + w_2x_1) = w_1^2\sigma_{11} + 2w_1w_2\sigma_{12} + w_2^2\sigma_{22}$

Subject to two constraints

  - Meet the return target $w_1r_1 + w_2r_2 \geq r$
  - and budget balance $w_1 + w_2 = 1$

Solution

- Start with the budget balance
- => $w_2 = (1- w_1)$ so replace
- Targe $r=w_1r_1+ w_2r_2$
- $\Leftrightarrow$ $r= w_1r_1+ (1- w_1) r_2$

  $\Rightarrow r =w_1r_1 -w_1 r_2 + r_2$

  $\Rightarrow r = w_1 (r_1 - r_2) + r_2 \Leftrightarrow w_1 = \frac{r-r_2}{(r_1-r_2)}$

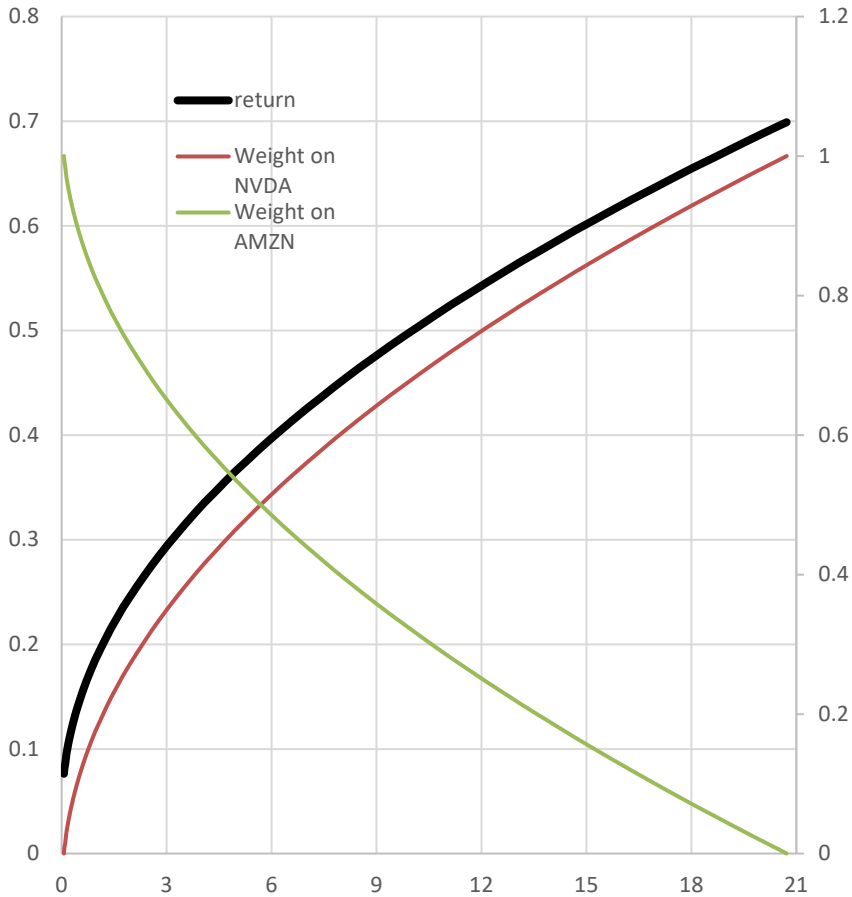
  $\Rightarrow$Given w_1, the portfolio has a return variance

$$Var(w_1x_1 + w_2x_1) = w_1^2\sigma_{11} + 2w_1w_2\sigma_{12} + w_2^2\sigma_{22}$$

Notice it's a quadratic function of w_1

More important each w_1 uniquely defines a return and variance. That gives us the portfolio variance

Mean Variance of Portfolios
No Short Sales 2-assets

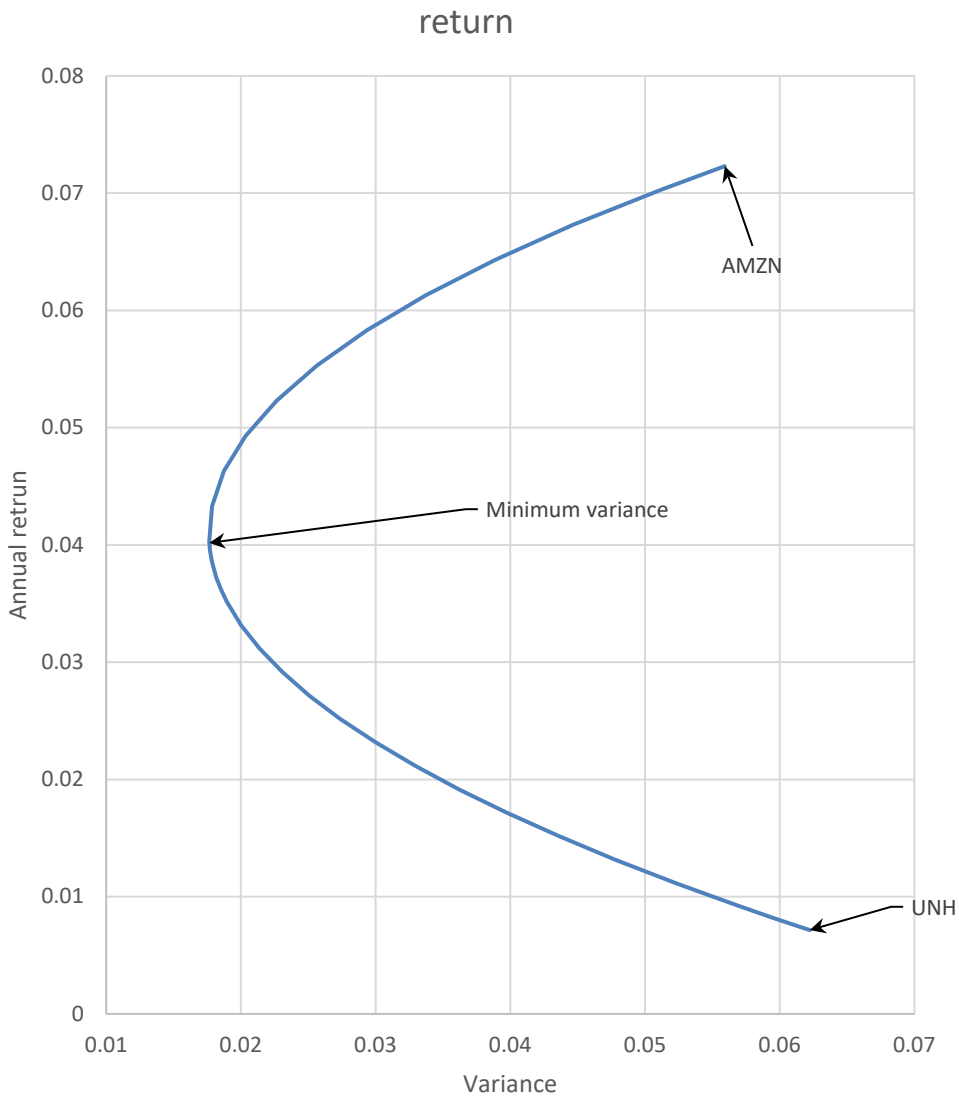


	Return statistics	
	NVDA	AMZN
Mean	0.699	0.076
Var	20.72	0.07
Covariance	0.98	

$$w_1 = \frac{r - r_2}{(r_1 - r_2)} = \frac{r - 0.76}{(0.62\ )}$$

$$\text{Var}\ (w_1) = w_1^2 20.72 + 2 w_1 w_2 0.07 + w_2^2 .098$$

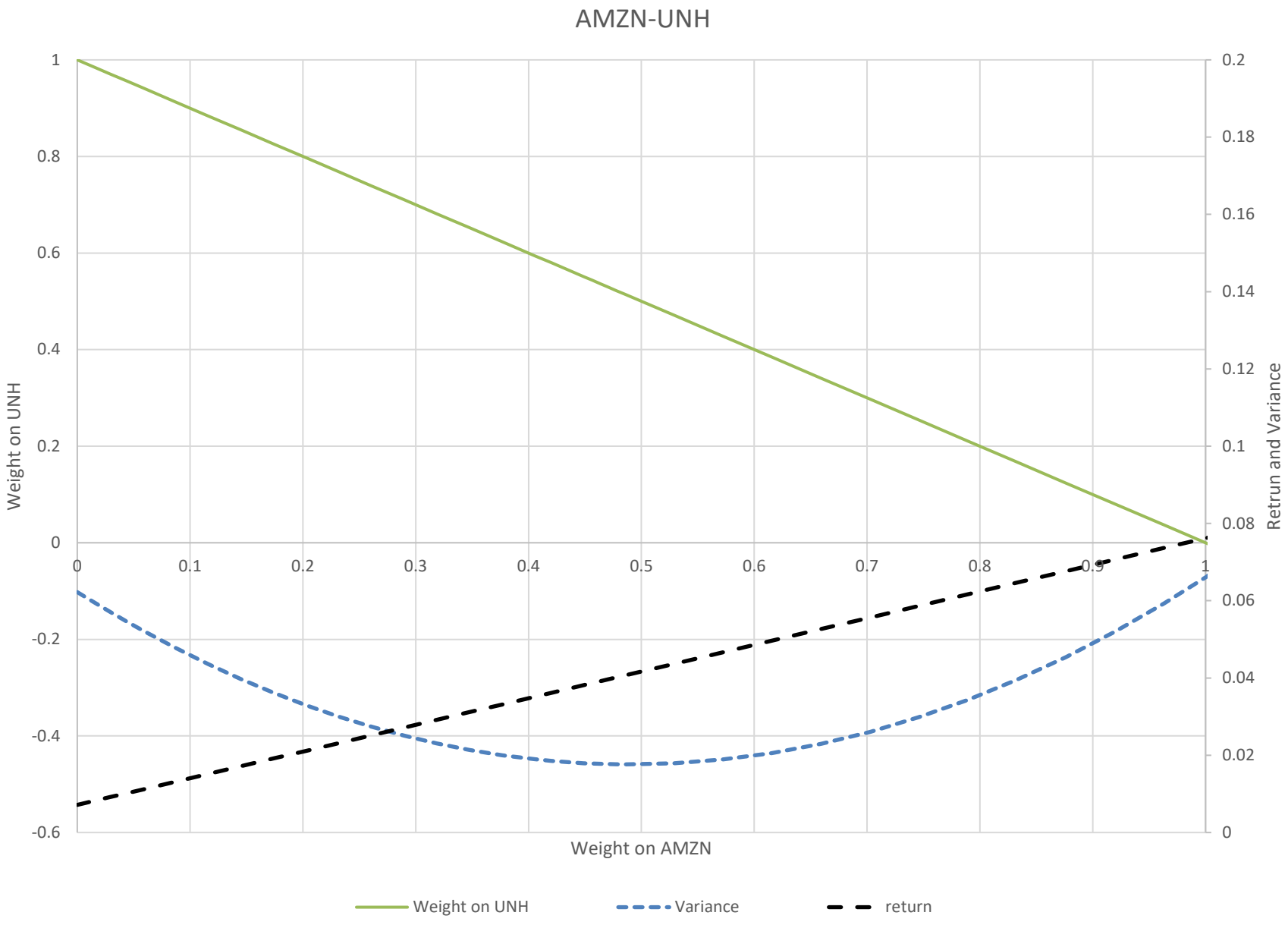
Mean Variance of Portfolios
No Short Sales 2-assets



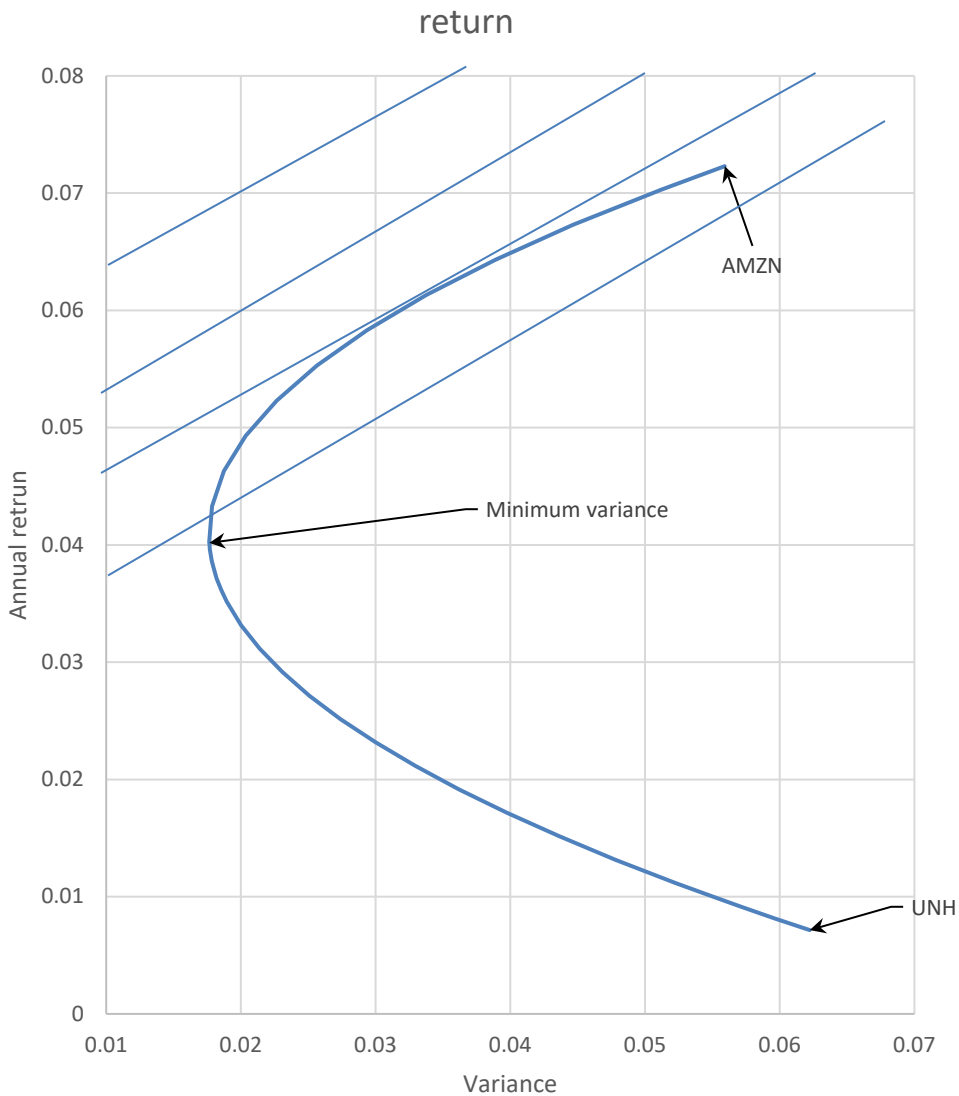
	Return statistics	
	AMZN	UNH
Mean	0.076	0.007
Var	0.066	0.062
Covariance	-0.029	

$$w_1 = \frac{r - r_2}{(r_1 - r_2)} = \frac{r - 0.007}{(0.069\ )}$$

$$\text{Var}\ (w_1) = w_1^2 0.066 - 2w_1 w_2 0.029 + w_2^2 .062$$


AMZN-UNH
Weight on UNH
Retrun and Variance
Weight on AMZN
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Weight on UNH
Variance
return

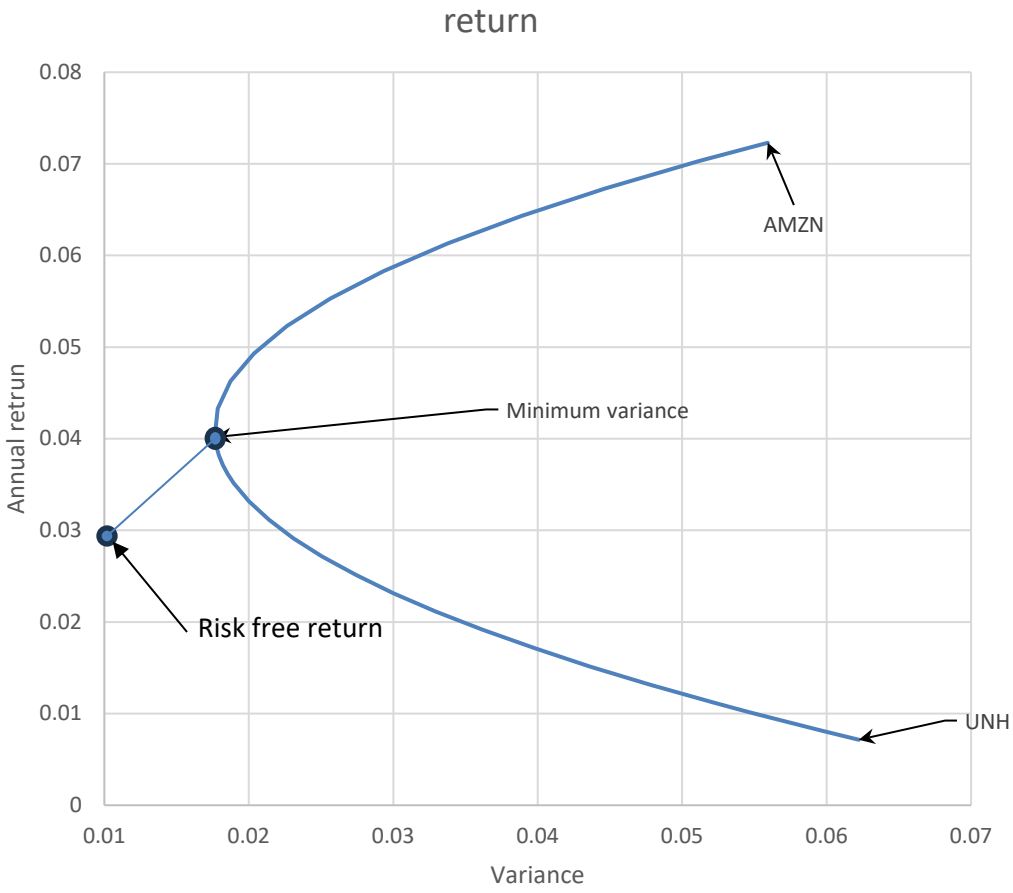
Mean Variance of Portfolios
No Short Sales 2-assets



- Return to mean-variance preferences
- Individual trades of return and risk with parameter b
- Best portfolio is the one that is on the frontier and such that the slope of the frontier is b. (here -1)
- Then you can find the certainty equivalent.

What if there is a risk-free asset

- You can do better
- Less variance than the min-variance portfolio
- Convex combination of riskless asset and min portfolio
- Then on frontier.



Sharpe Ratio S(x)

- What portfolio to mix with the riskless asset?
- S(x)=$\frac{r(x)-r_f}{Stdev(x)}$.
  - $r(x) - r_f$ is the rate at which you trade off return (the derivative of the linear part of the frontier
  - $Stdev(x)$ is the derivative of the horizontal axis
- The ratio tells us how much more return relative to the risk-free rate we get for the increased variance.
- Best Sharpe portfolio is max (highest net return conditional on variance).
- Notice that this value does depend on r_f
- S(x)=$\frac{r(x)}{Stdev(x)} - \frac{r_f}{stdev(x)}$. As the riskless return rises the optimal portfolio has more variance

Geometric approach

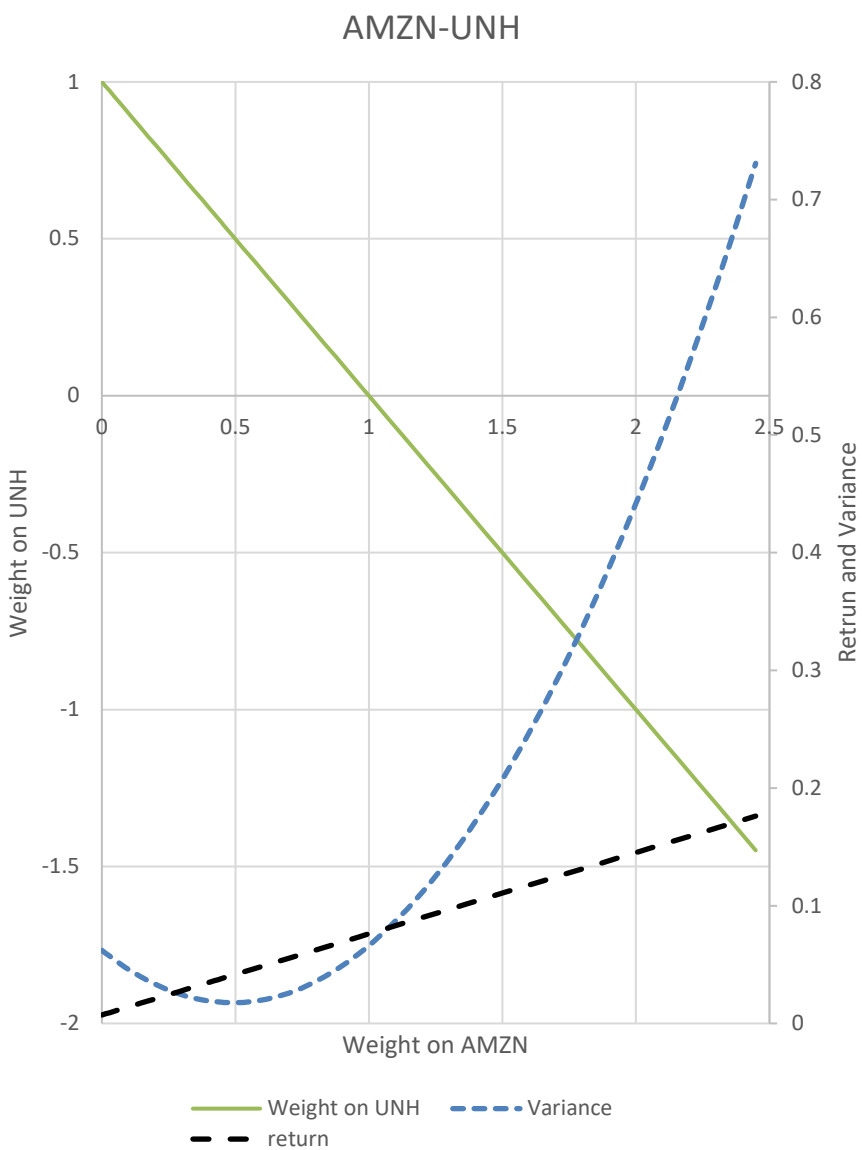
The dashed line has a slope r_x-r_f/ var(x). When the slope is tangent to the frontier we have a (feasible) maximum

Risk free asset

- If risk free-asset exists
- Portfolio separation theorem applies
- Optimal portfolios are combinations of the risk- free asset and the Sharpe optimal portfolio

- If no risk-free asset
- Investor must chose along the frontier

Short sales



Short sales
Need return of S2 is less than S1.
Going short: promising to sell S1 and adding more weight to S2 position.
You are 'borrowing' S2 to buy S1

<u>With risk free asset,</u>
Borrow at the risk-free rate (sell it short) go long on the Sharpe optimal portfolio.

<u>Without risk free asset</u>
Borrow S2 to buy more S1 (see graph)

Note this adds both return and variance
In either case

Leverage

- Now I borrow at a rate $r_f < r < r_s$
- This looks like the same problem.
- Except its not because of limited liability

Beyond 2 stocks

The data:
Note prices do not matter yet because you are figuring out proportions of your portfolio

x_1	x_2	…	x_i	…	x_n
r_1	r_2	…	r_i	…	r_n
σ_{11}	σ_{12}		σ_{1i}		σ_{1n}
σ_{21}	σ_{22}		σ_{2i}		σ_{2n}
….	….				
σ_{i1}	σ_{i2}		σ_{ii}		σ_{in}
…	…				
σ_{n1}	σ_{n2}		σ_{ni}		σ_{nn}

- The problem
- Find $W=\{w_1, w_2, \ldots w_i, \ldots w_n\}$
- That solve min Var(W) sbjt $r_W \geq r$
- So lets set this problem up

Beyond two stocks

- chose $w_1 \ldots w_n$ to minimize $$Var\left(\sum_{1}^{n} w_i x_i\right) = \sum_{i=1}^{n}\sum_{1}^{n} w_i w_j \sigma_{ij}$$
- Subject to two constraints
  - Meet the return target $$\sum_{i=1}^{n} w_i r_i \geq r$$
  - and budget balance $$\sum_{i=1}^{n} w_1 = 1$$
- Set up as a Lagrangean optimization
$$L = \sum_{i=1}^{n}\sum_{1}^{n} w_i w_j \sigma_{ij} + \lambda_1\left(r - \sum_{i=1}^{n} w_i r_i\right) + \lambda_2\left(1 - \sum_{i=1}^{n} w_1\right)$$
- There are now n+2 unknowns
$$(\mathrm{w}_1 \ldots \mathrm{w}_\mathrm{n}, \lambda_1, \lambda_2)$$
- So n+2 first order conditions. Because the Lagrangean is a polynomial of order 2, its FOC are n+2 linear equations with n+2 unknowns that can be solved numerically.

First order condition

- Take derivative with respect to the weights
- n FOCS $\boldsymbol{L} = \sum_{i=1}^{n}\sum_{1}^{n} \boldsymbol{w_i w_j \sigma_{ij}} + \boldsymbol{\lambda_1}\left(\boldsymbol{r} - \sum_{i=1}^{n} \boldsymbol{w_i r_i}\right) + \boldsymbol{\lambda_2}\left(\boldsymbol{1} - \sum_{i=1}^{n} \boldsymbol{w_1}\right)$

$$\frac{\partial L}{\partial w_i} = \sum_{j=1}^{n} w_j \sigma_{ij} - \lambda_1 r_i - \lambda_2$$

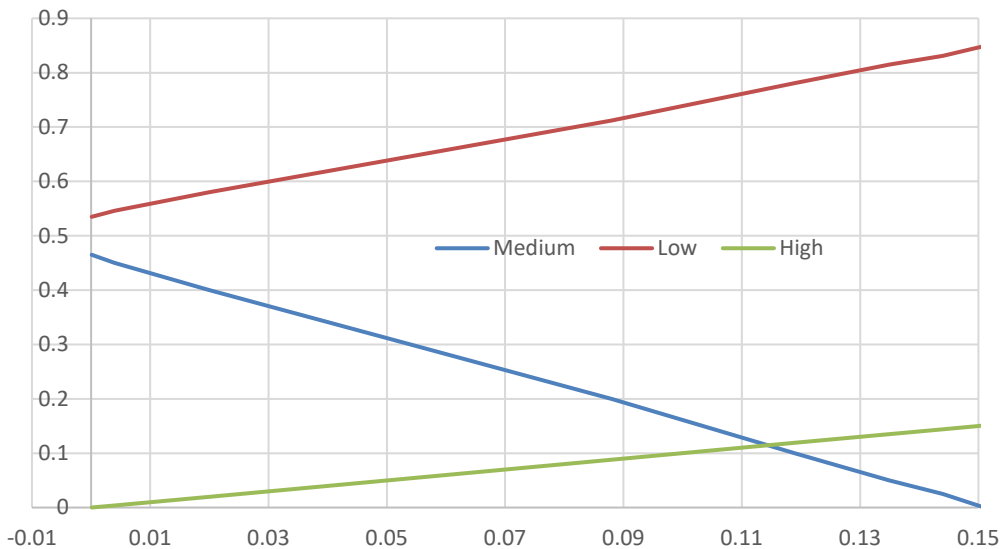
Two constraints

$$\frac{\partial L}{\lambda_1} = r - \sum_{i=1}^{n} w_i r_i$$

$$\frac{\partial L}{\lambda_2} = 1 - \sum_{i=1}^{n} w_i$$

Is this problem well behaved?

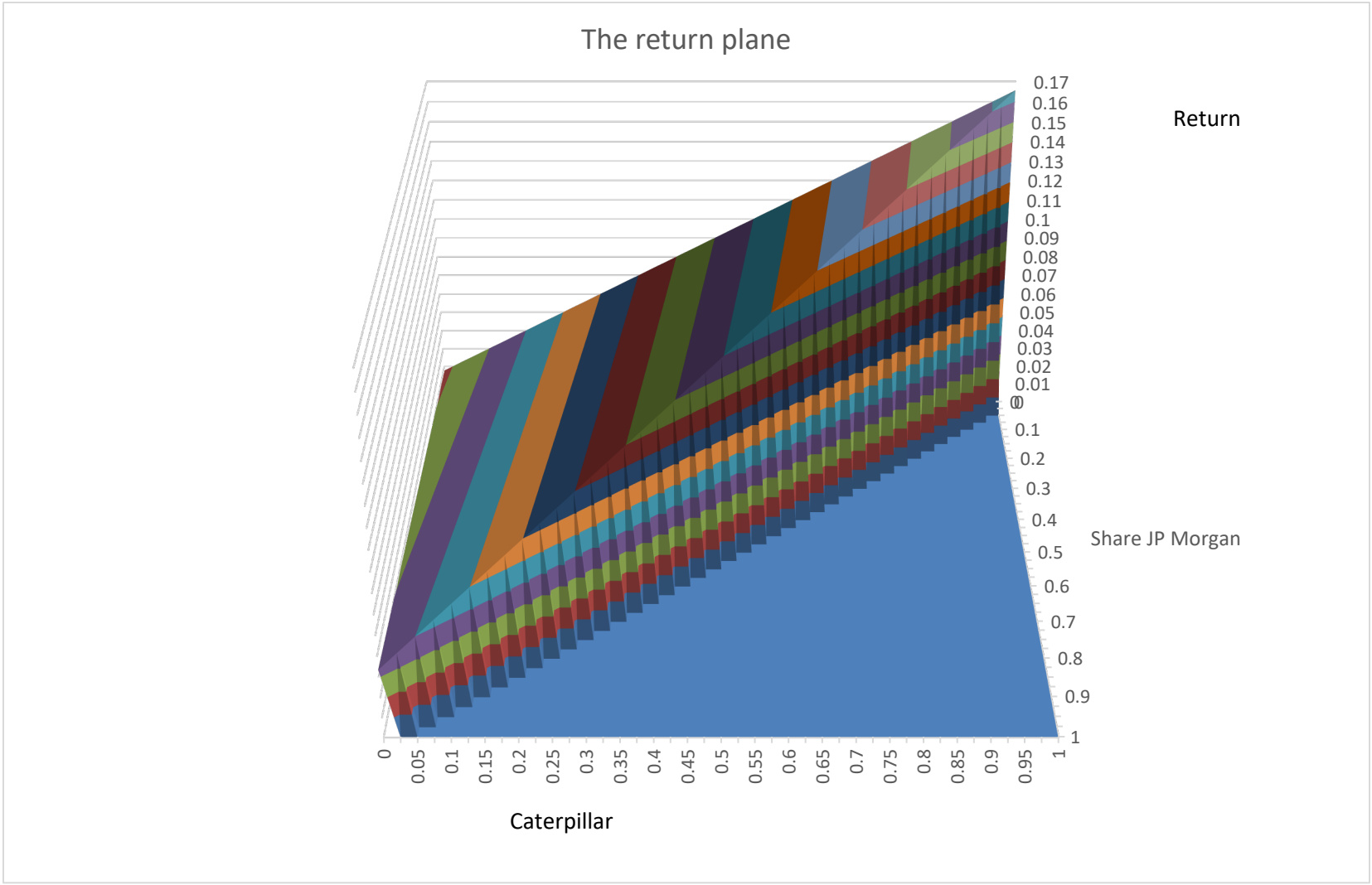
- Yes Intuition from three assets.
  - Start with a return between the medium return assets and the highest return asset. You are back to the two asset pb.
    - There is a unique pair of weights that give you that return.
  - Now for a return between the low-return asset and the medium return asset.
    - There is a unique portfolio that includes only the low and medium return asset. There is also a unique portfolio that includes only the low and high return asset.
    - The portfolios in between involve linear decreases in the share of the medium return asset and increases in the share the low return and high return asset



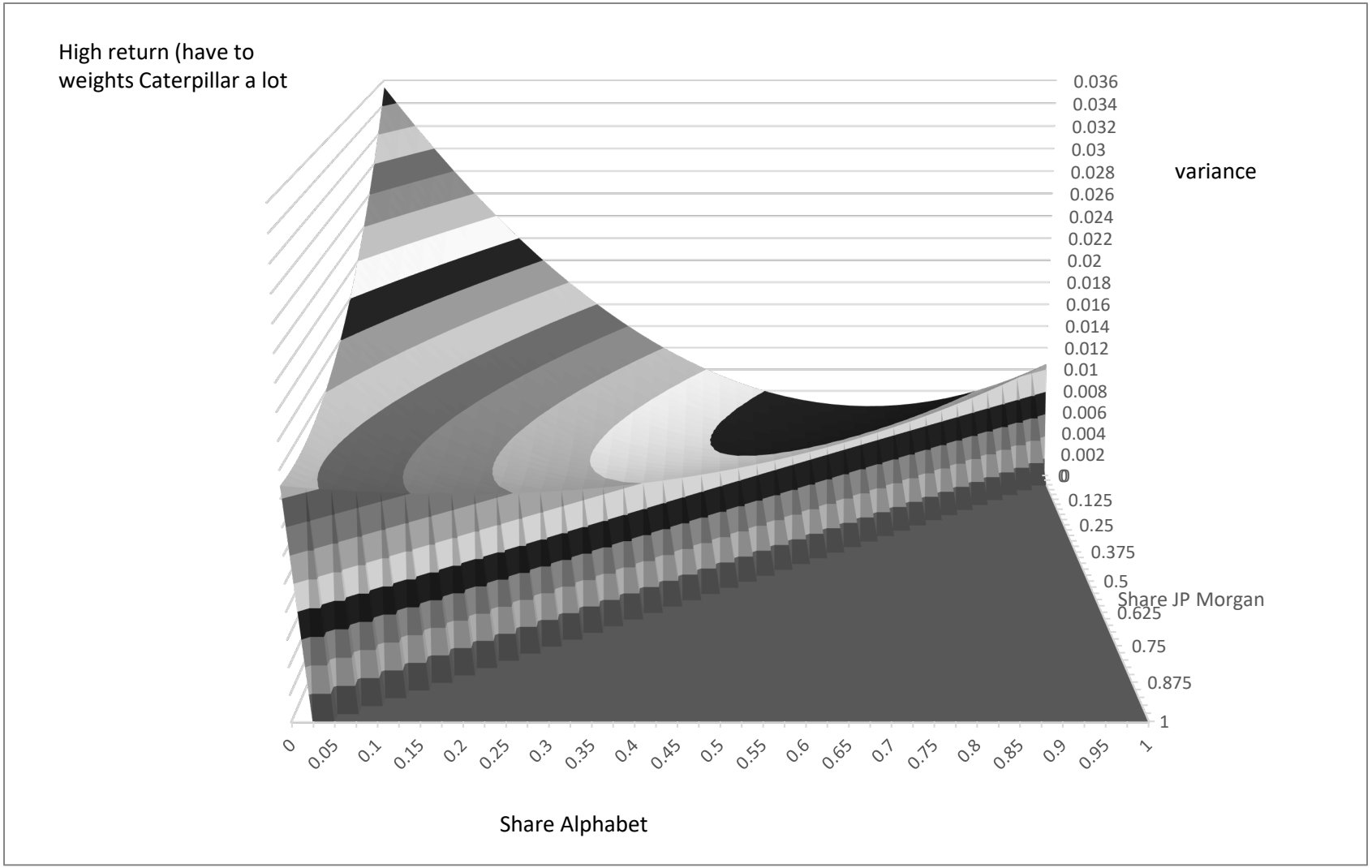
Is this problem well behaved?

- Now one can compute the variance across those portfolios
- It's a curve and it will have a unique min
- Induction works but you can get a computer to solve this really fast
- Problems
  - The key problem is the inputs (what are relevant forecasts of returns and variance)
  - Technical issue: when you have a lot of assets you are going to have some assets with very similar variance and covariances and if they are very correlated, they are close substitutes and that cause headaches because you may get multiple solutions.
  - What happens if two assets are very correlated (.99)
    - One has higher variance than the other
    - One has higher return that the other

Picking portfolios



The variance surface



Interpreting the return frontier

- If no short sales
  - best return is loading on the highest return asset (and its variance)
  - Other available optimal portfolios go from the min variance portfolio and are a concave curve up to the max.
- If there is a riskless asset (and No Short sale)
  - best return is loading on the highest return asset (and its variance)
  - Other available optimal portfolios start at the riskless asset and run on the tangency line to the frontier.
- If short sales (but no riskless asset)
  - Then you can have returns higher than those of the highest-return assets
- If short sales and riskless asset
  - Other available optimal portfolios start at the riskless asset and run on the tangency line to the frontier

Next Time

- What about the market?
- If people are min variance optimizers in choosing portfolios what does this do to prices?
- Need a model. CAPM